

Wellmere (grade 6 edition) — play-through review

Theme `seedbank_ms` · middle-school life science, grade 6: heredity and plant breeding · 9 days, 33 stops · reviewed 2026-08-21 by reading `books/seedbank-ms.yml` in full, working every board, and comparing against the parent `seedbank` .

Verdict

The junior edition that teaches the most, and the nine boards are the reason. All nine are correct and every one is a decision the breeding station has to make:

Board	Arithmetic	Idea
Counting a generation	$400 \times 0.75 = 300$ expected against 306 counted	a ratio as a prediction
Where a tonne of grain comes from	$11 \times 0.9 \approx 9.9$ tonnes out of the air	photosynthesis as mass
What the trial can actually claim	$11 \div 2 = 5.5$ times bigger in a dry season	an effect that depends on the year
Short is not simply worse	$8.6 \div 19.6 \approx 0.45$ against 0.33 for tall wheat	a ratio beats a total
Not the number of plants	$18 \div 120 \times 100 = 15\%$	effective population as a share
What one round returns	$7.4 + 0.3 \times 1.7 \approx 7.9$ t/ha	the breeder's equation
What was lost, and when	$30 \times 5 \div 100 = 1.5$ plants — one or two, often none	why a small grow-out loses things
What the catalogue says and what is there	$9,000 \times 0.33 \approx 3,000$; on 41,000 that is 13,500	scaling a sample to a collection
Twelve years, and what they are for	$2,352 \div 12 = 196$ mm against 118 this summer	an average against one year

Three of those are better teaching than the parent's equivalents. **"What was lost, and when"** — thirty plants at five per cent gives 1.5 expected, "one or two, often none", where two hundred plants would give ten — is genetic drift explained by an expected count, which is exactly how it should be explained and almost never is. **"Short is not simply worse"** turns a yield comparison into a ratio, which is the move the whole campaign is about. And **"Counting a generation"** predicts 300 and reports 306, so the ratio is evidence rather than a rule.

Answerable: 33/33. The mis-keyed DNA-marker stop this edition shared with the parent was fixed earlier in this review; both are green. **Sense:** Excellent. Three people want the same ground and none of them is wrong. **Level:** Right. `questionLoad` reports 4 of 33 demanding stops (12%), and the campaign computes 4 of 4 syllabus relations — one of only four junior editions at 100%. **Fun:** Good, and the rings are the best-conceived place in the junior set: the geography *is* the genetics.

Implemented since this review

- **WM-01's arithmetic** (16 of 400, not 11) in this edition too.
- **SM-02** `dayNoun` .
- Numeral damage.

Everything else in the findings table below is unchanged, and `npm run check` is green on this theme.

Findings

ID	Sev	Where	Issue	Proposed fix
SM-01	PLAN	Six of nine days author 4 stops	<code>per-day</code> is <code>[4,4,4,3,3,4,4,4,3]</code> . Six of nine days at the loop's maximum, second only to <code>planetary_defense_ms</code> , and it means six of nine days carry no callback. Thirty-three stops in nine days is 3.7 a day: the content wants ten days.	Reclassified for the same reason as <code>planetary_defense_ms</code> 's PM-02 — six four-stop days is a real load problem, and it is not on its own why this edition has no callback. See <code>_plan.pdf</code> .
SM-02	WORTH	<code>themes/seedbank_ms/theme.js</code>	No <code>dayNoun</code> . Three junior editions are missing it — this, <code>aftershock_ms</code> and <code>icecore_ms</code> . The clock here is "sowing starts in three weeks" and the plan card prints "Day N".	One line. Cross-campaign §4.
SM-03	WORTH	<code>engine/dev/concept-debt.json</code> , 5 rows	Five ordering rows, no <code>takesAsRead</code> at grade 6. CLAUDE.md records that moving Wellmere's bagging stop to day 1 made <code>jargonDepth</code> fire, because that card defines <i>Pollen</i> with <i>Chromosome</i> , unseen until day 4 — so this campaign has already had one move rejected on a vocabulary ground, which is worth noting in the debt file so it is not retried. Three of the five point at <i>variety inside a collection, and why losing it is hard to undo</i> , which is what "What was lost, and when" teaches.	Claim it explicitly at that stop rather than moving anything. One <code>concept:</code> line may clear three rows.
SM-04	WORTH	Zero — Review variants	No spaced retrieval. Cross-campaign §9. This edition has the strongest case of the twelve: the	Three or four variants, best added with SM-01's extra day.

drift idea on day 6 or 7 is the one a child will not hold from a single serving.

SM- TASTE 05	Numeral damage, 3 occurrences	"1 field", "1 door", "1 plant". Minimal — the parent has twenty.	Editorial pass. Cross-campaign \$1.
SM- TASTE 06	The opening card	Names both Dr. Chiara Volpe and Dr. Nasrin Qureshi with their jobs and their positions — " <i>wants a new wheat sent out to farmers this year</i> " against " <i>the oldest samples must be grown again first, while enough of them still come up</i> ". Both sides of the argument, on the card, at grade 6. Recording it as the model.	Nothing. This is what the eleven campaigns whose opening cards name nobody should look like.

What the derivation did well

- **It kept the three-way argument over ground**, and it kept all three people right.
- **Drift as an expected count.** $30 \times 5 \div 100 = 1.5$ is the best single idea in the junior set.
- **The label.** "WM-712 went to the merchant with its weakness written on the label" survives into the ending, which is the campaign's ethical position and a sixth grader can hold it.
- **"The records still disagree with the markers on 60 samples."** An unfinished item that is a number.

Opening and closing

Opening: 108 words, both sides named with positions. Best junior opening in the set alongside `icecore_ms`.

Closing: three paragraphs. "What is in the ground this year is what you..." — the last paragraph is addressed to the player and gives the credit in the campaign's own terms.

Warm-ups

Authored for this edition. This is a two-tier site, so `trial-far` is present. No findings.